

$$\mathcal{L} = \sqrt{g} \left[ \frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) + \frac{1}{12} R \phi^2 \right]$$

$$\psi \phi = \lambda \phi, \quad \delta g_{\mu\nu} = -2\lambda g_{\mu\nu}$$

$$\Leftrightarrow \phi \rightarrow e^{\lambda_0} \phi \quad g_{\mu\nu} \rightarrow e^{2\lambda_0} g_{\mu\nu}$$

If we fix the Gauge  $\phi := \frac{\sqrt{6}}{R}$   
we recover the Einstein-Hilbert Action  
for pure gravity:

$$\boxed{S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} R}$$

Conformal Killing Eqs.:

$$x^\mu \mapsto x^\mu + \xi^\mu \in \mathbb{R}, \quad \epsilon \ll 1$$

$$\Rightarrow \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \frac{2}{D} (\eta_{\mu\nu} \partial_\rho \xi^\rho) = 0$$

General Sol<sup>n</sup>:

$$\xi^\mu(x) = a^\mu + \lambda^{\mu\nu} x_\nu + \lambda_D x^\mu + (x^2 \lambda_\mu^\mu - 2x^\mu \lambda_\mu^\mu)$$

$$1) \quad \frac{D(D-1)}{2} \quad 1 \quad D$$

$$2D+1 + \frac{D(D-1)}{2} = \frac{4D+2+D^2-D}{2} = \frac{3D+2+D^2}{2}$$

$$\dim(\text{Conf}(M^D)) = \frac{(D+1)(D+2)}{2}$$

which specifies the number of generators  
of  $\text{Conf}(M^D)$ . In Particular the Poincaré

group is a subgroup.

$$\text{ISO}^{\uparrow}(M) \subset \text{Conf}(M^D)$$

Case 1  $\boxed{S(x) = \eta_{\mu\nu} x^\mu x^\nu + \lambda_D x^\mu x_\mu + \lambda^{\mu\nu} x_\mu x_\nu + a^\mu x_\mu}$

Generators:  $[O(\epsilon)] := a P_\mu + \frac{1}{2} \lambda M_{\mu\nu} + \lambda_0 D + \lambda_i K_i$

$\rightarrow$  For  $[D, P_\mu] = -P_\mu$   
 $[D, K_\mu] = -K_\mu$

Notes use  $+P_\mu$ , which is 'wrong'

We can embed the Conformal group into  $SO(D, 2)$ , the isometry group of  $AdS_{D+2}$  -

$M^{\mu\nu} := \begin{pmatrix} \hat{M}^{\mu\nu} & \frac{1}{2}(P^\mu K^\nu - P^\nu K^\mu) & \frac{1}{2}(P^\mu + K^\mu) \\ -\frac{1}{2}(P^\mu - K^\mu) & 0 & -D \\ -\frac{1}{2}(P^\nu + K^\nu) & D & 0 \end{pmatrix}$

where the metric on  $SO(D, 2)$  is  $(S(D, 1+1)) : \hat{M} := \text{diag}(-1, 1, \dots, 1, -1)$

\*Note  $\boxed{D = x^\rho \partial_\rho}$   
 $K_\mu = x^2 \partial_\mu - 2x_\mu x^\nu \partial_\nu$

Conformal Trans. on fields:

Above, we really should have  $\mu \rightarrow a$  for elements/Lorentz on  $M^D$ .

{ "orbital-part" : Acts on spacetime pt  $x^\mu$   
 "Spin part" which acts on the rep index.  
 $D, K_\mu$ : Also have "orbital parts" as well



as "intrinsic parts", which we denote by  $K_D^i(\phi)$ ,  $K_M^i(\phi)$

The global Nature is absorbed into  $\xi^\mu(x)$ .

$$\delta_c \phi^i(x) \equiv \xi^\mu(x) \partial_\mu \phi^i(x) - \frac{1}{2} \Lambda^{\mu\nu}(x) (M_{\mu\nu})^i_j \phi^j(x) + \Lambda_D(x) K_D^i(\phi)(x) + \lambda_K^\mu K_M^i(\phi)(x)$$

where

$$\Lambda_{\mu\nu}(x) := \partial_\mu \xi_\nu - \partial_\nu \xi_\mu = \lambda_{\mu\nu} - 2/x_\mu \lambda_{\nu\mu}$$

$$\Lambda_D(x) := \frac{1}{2} \partial_\rho \xi^\rho = \lambda_D - 2x \cdot \lambda_K$$

Q: where does  $\Lambda_D(x)$  come from?

$$* \quad x^\mu \mapsto x^\mu + \xi^\mu \equiv x'^\mu$$

$$\Rightarrow g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}(x)$$

$$\Rightarrow \delta(g^2) = 2 \Lambda_D dg^2,$$

$$\boxed{\Lambda_D = \lambda_D - 2(x \cdot \lambda_K)}$$

ex:  $\phi$  w/ intrinsic dilation  $K_D(\phi) \equiv \omega \phi$

Then  $\partial_\mu \phi(x)$  transforms as a field w/ conformal weight  $(\omega+1)$ .

$$\text{Proof: } \partial_\mu (\delta_c \phi) = \dots + (\omega+1) \partial_\mu \phi$$

by using

$$[D, P_\mu] = -P_\mu, [D, K_\mu] = -K_\mu$$

ex: Maxwell in  $D=4$  is invariant under Dilations & SCT's.

• Use  $\delta A_\mu$ , which has a Conf. weight of 2

so  $KD/M = A_M$  and so  $\gamma$  no extra  $U(1)$

$$K_M^\nu = 0, \quad \Delta_M = (\lambda_0 - 2x \cdot \lambda_n)$$

So we need  $(M_{[uv]}) \rho^\sigma = 2 \Omega_{\rho\sigma} S_{uv}^\sigma$

$$\Rightarrow \delta E_{\mu\nu} = \int d^4x F_{\mu\nu} + 2 \Delta_{\mu\nu}^\rho F_{\rho\sigma} + 2 \Delta_0 F_{\mu\nu}$$

So  $F$ : has a conformal weight of  $\underline{2}$

\* Now work towards a Gauged Conf. Symm.

->  $\phi^i(x)$  lives on a "dynamical" spacetime m.f.d. determined by the frame field  $e_\mu^a(x)$

$$\mu \rightarrow a$$

Def: World Scalar / Coord scalars: Fields w/ only coordinate scalars.

$$\boxed{\xi^a(x) := e_\mu^a(x) \xi^\mu(x)}$$

Standard Gauge trans' of the Conf. algebra:

$$\delta(\epsilon) = \epsilon^A T_A \equiv \frac{1}{2} \lambda^{ab}(x) M_{[ab]} + \lambda_0(x) D + \lambda_n^\alpha(x) K_\alpha$$

-> We absorb all the individual trans.

of  $\xi^\mu(x) = a^\mu + \lambda^{\mu\nu} x_\nu + \lambda_0 x^\mu + (x^2 \lambda_n^\mu - 2x^\mu(x \cdot \lambda_n))$  into the single parameter  $\xi^\mu$ .

-> Similarly, we absorb  $(\Delta_n)_\alpha^\mu(x)$  into the Lorentz parameters  $\lambda^{ab}$  and  $\Delta_0(x)$  into  $\lambda_0(x)$ .



\* Note, this is simply a change of basis of  $\langle \Delta N A(M^D) \rangle$  which are spanned by the above generators  $\{T_A\}$ .  
 $\{M_{ab}, D, K_a\}$  satisfy the  $\text{Conf}(M^D)$  commutation relations

- Replace translations by gct's:  
 $\delta_{\text{gct}} \phi(x) := \xi^a(x) e_a^\mu \partial_\mu \phi(x) = \xi^a(x) \partial_a \phi$   
 We established,  $\xi^a$  &  $\lambda^{ab}$  formed a basis for the Gauge parameters of the Poincaré group.

\* Local CFT's of fields that are coord. scalars can thus be written as

$$\delta_c \phi^i = \xi^a(x) \partial_a \phi^i - \frac{1}{2} \lambda^{ab}(x) (M_{[ab]})^i_j \phi^j(x) + \lambda_D(x) \Delta(\phi)(x) + \lambda_a^i K_a^i(\phi)(x)$$

(i) Covariant gen. coord. trans:

As just stated, the first term is simply  $\delta_{\text{gct}} \phi^i = \xi^a(x) \partial_a \phi^i(x)$

More, generally, we have the Covariant gct - c gct

$$\delta_{\text{cgct}}(\xi) = \delta_{\text{gct}}(\xi) - \delta(\xi^\mu B_\mu)$$

$$\text{So get } (\xi) \phi^i = \xi^\mu \partial_\mu \phi^i + (\xi^\mu A_\mu^A) (T_A)^i_j \phi^j$$

$$= \xi^\mu D_\mu \phi = \xi^a D_a \phi$$

$$D_a \phi = e_a^\mu D_\mu \phi$$

$$D_\mu \phi = \partial_\mu \phi - B_\mu^A (T_A \phi)$$

In particular, recall:

$$(T_A \phi)^i := - (t_A)^i_j \phi^j \quad (\star)$$

### (ii) Lorentz trans.:

- Encoded by  $(M_{ab})^i_j$ .

(a) Scalars:  $(M_{ab})^i_j \in \mathbb{C}$

$$\text{So } (a \cdot b) \stackrel{!}{=} ba \Rightarrow M_{ab} = 0$$

(b) Spinors:  $i$  &  $j$  are spinor (Spinor) indices, which are generally not written.

$$M_{ab} = \frac{1}{2} \gamma_{ab}$$

$$\hookrightarrow \exp\left(\frac{1}{2} \lambda^{ab} M_{ab}\right) \psi = \psi - \frac{1}{4} \lambda^{ab} \gamma_{ab} \psi$$

$$\delta \psi = - \frac{1}{4} \lambda^{ab} \gamma_{ab} \psi$$

$$(c) \text{ Vectors: } (M_{ab})^c_d = 2 \delta_{[a} \eta_{b]d}$$

$$\Rightarrow \delta V^a = - \lambda^{ab} V_b$$

### (iii) Dilations

Specified by  $R_D^i(\lambda)(x)$ . We will, for now, only consider Primary fields of conformal weight  $h$  (Previously,



$$K_D^i(\phi)(x) := h \phi$$

$$\Rightarrow \delta \phi^i = \omega \lambda_D \phi^i$$

→ For a vector  $V^\mu$ , this has a conf. weight of 1. The metric,  $g_{\mu\nu}$ , has a conformal weight of -2.

• So  $V^\mu$  will have conf. weights  $n$  while elements of  $\omega \lambda_D V^\mu$  will have  $-n$

(iv) Special Conf. Trans.:  $K_a^i(\phi)(x)$

$$\delta_{\text{SCT}} \phi^i = \lambda^a_\mu K_a^i(\phi) \equiv \lambda^a_\mu \chi_a^i$$

where  $\chi_a^i$  may be another field of the system or a derivative of another field.

In effect,  $D$  acts as an op w/  $\phi^i$ , a primary field, the eigen states of  $D$ . Why?

$$M_D \phi^i = e^{\lambda_D D} \phi^i = \phi^i + \lambda_D D \phi^i$$

$$\text{But } \lambda_D (D \phi^i) \neq h \lambda_D \phi^i$$

With the above box, we find, along w/  $[D, K_\mu] = -K_\mu$

$$D(K_\mu \phi^i) = K_\mu (D \phi^i) - K_\mu \phi^i = (h - 1) K_\mu \phi^i$$

\* We conclude that acting on a field  
w/  $K_\mu$  lowers the weight of the  
primary field by 1

$$K_\mu : \phi \mapsto K_\mu \phi$$

$$(h) \mapsto (h-1)$$

\* If the theory does not contain a  
field w/ such a weight then  $\phi^i$   
has vanishing SCT.

• Note if  $D K_\mu \phi^i = 0$  then  $\phi^i$   
has weight 1 or  $K_\mu \phi^i = 0$ .

Claim:  $e_\mu^a$  has conf. weight -1

[iff  $\phi$  has (c.w.)  $h$ , then

$$D_a \phi \equiv e_a^\mu (\partial_\mu - h b_\mu) \phi \text{ has (c.w.) } h+1$$

where  $b_\mu$  is the Gauge field transf.  
under dilations as  $\delta_D b_\mu = \partial_\mu \lambda_D$

Proof: (i)  $e_\mu^a \eta_{ab} e_\nu^b = g_{\mu\nu}$

$$\Rightarrow \delta_D (e_\mu^a e_\nu^a) = \delta g_{\mu\nu} = -2\lambda_D g_{\mu\nu}$$

$$\Rightarrow 2 \delta_D (e_\mu^a \eta_{ab} e_\nu^b) = -2\lambda_D e_\mu^a \eta_{ab} e_\nu^b$$

$$\therefore \delta_D e_\mu^a = -\lambda_D e_\mu^a \Rightarrow \boxed{h=1}$$

$$(ii) \delta_D (D_a \phi) = \delta_D (e_a^\mu \partial_\mu \phi) = h \delta_D (b_\mu \phi)$$

So note, this is an abelian gauge

$$\delta_D e_a^\mu \partial_\mu \phi + e_a^\mu \delta_D (\partial_\mu \phi) = h \delta_D (b_\mu \phi)$$

$$(\delta_D e_a^\mu) \partial_\mu \phi + e_a^\mu \delta_D (\partial_\mu \phi) = h \delta_D (b_\mu \phi)$$



$$\begin{aligned}
&= +\lambda_0 D_a \phi + e_a^\mu (\hbar+1) \partial_\mu \phi \lambda_0 - \hbar \partial_\mu \lambda_0 \phi \\
&\quad - \hbar b_\mu \lambda_0 \hbar \phi \\
&= \hbar \lambda_0 e_a^\mu (\partial_\mu \phi - \hbar b_\mu \phi) + \lambda_0 D_a \phi \\
&\quad + e_a^\mu \partial_\mu \phi \lambda - \hbar (\partial_\mu \lambda_0) \phi e_a^\mu \\
&= (\hbar+1) \lambda_0 D_a \phi + \cancel{e_a^\mu (\partial_\mu \phi \lambda)} + \\
&\Rightarrow \text{Not sure why extra terms do not} \\
&\quad \text{vanish}
\end{aligned}$$

### The Gauge Fields & Constraints:

→ Introduce gauge fields for the Conf alg. We list the generators, Gauge parameters, & Gauge field

| $(T_A)$<br>Gen. | $P_a$     | $M_{ab}$          | $D$         | $K_a$         |
|-----------------|-----------|-------------------|-------------|---------------|
| $\Theta_A(x):$  | $\xi^a$   | $\lambda^{ab}$    | $\lambda_0$ | $\lambda^a_k$ |
| $B_\mu^A:$      | $e_\mu^a$ | $\omega_\mu^{ab}$ | $b_\mu$     | $f_\mu^a$     |

$$\begin{aligned}
&\delta(\Theta(x)) \phi^B \equiv \Theta^A(x) (T_A \phi^B) \\
&\Rightarrow D_a \phi^A \equiv \partial_a \phi^A - \Theta^c(x) B_\mu^B f_{BC}^A
\end{aligned}$$

This will lead us to a theory of gravity.

As usual,  $P_a$  are gauged by the frame fields  $e_\mu^a(x)$  and play a

Special Role  $m$ . They will be dist. from the 'standard' twist, generated by  $\{M_{ab}, D, K_a\}$ .

Recall that,  $\delta(\epsilon) B_\mu^A \equiv \partial_\mu \epsilon^A + \epsilon^C B_\mu^{BC} f_{BC}^A$   
as above. The structure constants  $f$   
can be read off of the table in text  
(12.2 - pg 218)

These "general" results are then:

$$\begin{aligned}\delta e_\mu^a &= -\lambda^{ab} e_{\mu ab} - \lambda_D e_\mu^a \\ \delta \omega_\mu^{ab} &= \partial_\mu \lambda^{ab} + 2 \omega_\mu^c [{}^a \lambda^b]_c - 4 \lambda_\mu^{[a} e_r^{b]} \\ \delta b_\mu &= \partial_\mu \lambda_D + 2 \lambda_\mu^a e_{\mu a} \\ \delta f_\mu^a &= \partial_\mu \lambda_\mu^a - b_\mu \lambda_\mu^a + \omega_\mu^{ab} \lambda_{\mu b} - \lambda^{ab} f_{\mu ab} \\ &\quad + \lambda_D f_\mu^a\end{aligned}$$

Note, we will need structure constants  
for  $K_\mu$  &  $D$  terms

$$[D, P_\mu] = -P_\mu \rightarrow f_{\mu a}^a = -\delta_\mu^a (K_a)$$

etc. ...

Proposed:  $\delta e_\mu^a = \partial_\mu \xi^a + \xi^B B_\mu^{BC} f_{BC}^a$  ???

Note, we end up w/  $\partial_\mu \xi^a = 0$

(Not sure why, if we have gauged  
these symm then  $\partial_\mu \xi^a$  should

not vanish if  $\partial \xi^a(x) = e_\mu^a \partial \xi^a(x)$ )



$$\begin{aligned}
 \therefore \text{We get } \lambda^{cd} e_{\mu}^f & \quad f_{f[cd]}^a \\
 &= \lambda^{cd} e_{\mu}^f (2 \eta_{f[c} \delta_{d]}^a) \\
 &= \lambda^{cd} e_{\mu}^f (\eta_{fc} \delta_d^a - \eta_{fd} \delta_c^a) \\
 &= \lambda^{cd} e_{\mu c} \cdot = \lambda^{ad} e_{\mu d} \\
 &= -2 \lambda^{ab} e_{\mu b}
 \end{aligned}$$

But, this is off by a factor of  $\frac{1}{2}$ .

Not too sure about this.

Second term:

$$\lambda_D^? e_{\mu}^B f_{B?}^A \equiv \lambda_D e_{\mu}^B (-\delta_B^A)$$

$\therefore$  From  $[D, P_a] = -P_a$  we have

$$\boxed{f_a^b \equiv -\delta_a^b}$$

$\rightarrow$  These are the "structure" const. which we use when dealing w/ commutators of

$$D: [D, P_a] = -f_a^b P_b \text{ etc. } \square$$

\* Note, we obtain new terms from  $[P_a, K_b]$  which yield non-zero structure const.

Q: How do we distinguish between  $f_{ab}^c = 0$  of  $[P, P] = 0$  and  $[P, K] \neq 0$ ? Perhaps, we label these points of  $\mathcal{H}$  by  $\tilde{a}$  and  $a$  on  $P \Rightarrow f_{a\tilde{a}}^?$

\* Note, In the above,  $e$  has weight  $-1$   
 $f_m^a: +1$ , all others  $0$

Recall, that these transformations

must close

$$[\delta(\epsilon_1), \delta(\epsilon_2)] = \delta(\epsilon^3) \quad | \quad \epsilon^3 \equiv \epsilon_2^B \epsilon_1^A f_{AB}^C$$

ex:  $[\delta_1, \delta_2] e_\mu^a$

$$= \delta_1 (-\lambda_2^{ab} e_{\mu b} - \lambda_D^2 e_\mu^a)$$

$$= -\lambda_2^{ab} \eta_{bc} (-\lambda_1^{cd} e_{\mu d} - \lambda_D^1 e_\mu^c)$$

$$- \lambda_D^2 (-\lambda_1^{ab} e_{b\mu} - \lambda_D^1 e_\mu^a)$$

$$= + \lambda_2^{ab} \eta_{bc} \lambda_1^{cd} e_{\mu d} + \lambda_2^{ab} \eta_{bc} \lambda_D^1 e_\mu^c$$

$$+ \lambda_D^2 \lambda_1^{ab} e_{b\mu} + \lambda_D^2 \lambda_D^1 e_\mu^a$$

$$= \lambda_2^a \lambda_1^{bc} e_{\mu b} + \lambda_2^a \lambda_D^1 e_{\mu b}$$

$$+ \lambda_D^2 \lambda_1^{ab} e_{b\mu} + \lambda_D^2 \lambda_D^1 e_\mu^a$$

$$= (\lambda_2^a \lambda_1^{cb} + \lambda_2^a \lambda_D^1 + \lambda_D^2 \lambda_1^{ab} + \lambda_D^2 \lambda_D^1 \eta^{ab}) e_{\mu b}$$

The only independent field in Einstein gravity is the frame field, so we do not want all the extra D.O.F.

$\leadsto \omega_\mu^{ab}$  is a composite field, determined by a constraint of the translation curvature

$$R_{\mu\nu}(P^a),$$

$$\text{Recall, } [D_\mu, D_\nu] = -\delta R_{\mu\nu}$$

$$R_{\mu\nu}^A = 2 \partial_\mu B_\nu^A - B_\nu^C B_\mu^B f_{BC}^A$$



$$\Rightarrow R_{\mu\nu}(P^a) = 2\partial_{[\mu} e_{\nu]}^a + 2\omega_{\mu}^{ab} e_{\nu]b} - \frac{1}{2}\epsilon_{\mu\nu}^{\alpha\beta}\gamma_{\alpha}\gamma_{\beta}$$

→ Need a new value for  $\omega_{\mu}^{ab}$  expanded to include the new conf. fields

↳ One new term to the  $R_{\mu\nu}(P^a)$  above from  $[D, P_a] = -P_a$

• We also note, that the translational curvature vanishes via the condition  $\nabla_{\mu} e_{\nu}^a = 0$ .

$$\Rightarrow R_{\mu\nu}(P^a) \equiv 2(\partial_{[\mu} + b_{[\mu}) e_{\nu]}^a + 2\omega_{\mu}^{ab} e_{\nu]b} = 0$$

→ This term is for connections w/ torsion which we associate to the SUSY But, so far we have not included SUSY into the theory. It is purely a Conf. extended theory.

\* Solving this for  $\omega_{\mu}^{ab}$ :

$$\omega_{\mu}^{ab}(e, b) := \omega_{\mu}^{ab}(e) + 2e_{\mu}^{[a} e^{b]\nu} b_{\nu}$$

$$\omega_{\mu}^{ab}(e) \equiv 2e^{\nu c} \partial_{[\mu} e_{\nu]}^{b]} - e^{\nu a} e^{b]\sigma} e_{\mu c} \partial_{\sigma} e_{\nu}^c$$

Proof:

$$\partial_{[\mu} e_{\nu]}^a + b_{[\mu} e_{\nu]}^a + \omega_{\mu}^{ab} e_{\nu]b} = 0$$

$$\omega_{\mu}^{ab} e_{\nu]b} = -\partial_{[\mu} e_{\nu]}^a - b_{[\mu} e_{\nu]}^a$$

$$(\omega_{\mu}^{ab} e_{\nu]b} - \omega_{\nu}^{ab} e_{\mu]b}) = -2\partial_{[\mu} e_{\nu]}^a - 2b_{[\mu} e_{\nu]}^a$$

$$e_{\mu}^c e_{\nu}^d ( ) = -2e_{\mu}^c e_{\nu}^d (\partial_{[\mu} e_{\nu]}^a + b_{[\mu} e_{\nu]}^a)$$

$$\Rightarrow e_c^\mu \omega_\mu^{ab} \eta_{bc} \partial_d - \omega_\nu^{ab} \eta_{bc} e_d^\nu \partial_c = \text{RHS}$$

$$e_c^\mu \omega_\mu^{ab} \eta_{bd} - \omega_\nu^{ab} \eta_{bc} e_d^\nu = \text{RHS}$$

$$\Rightarrow \omega_\mu^{ab} (e_c^\mu \eta_{bd} - e_d^\mu \eta_{bc}) = \text{RHS}$$

$$\omega_\mu^{ab} ( ) = -2 e_c^\mu e_d^\nu (\partial_\mu e_\nu^a + b_\mu e_\nu^a) \quad (2)$$

$$\left\{ \begin{aligned} \text{RHS} &= -e_c^\mu e_d^\nu (b_\mu e_\nu^a - b_\nu e_\mu^a) \\ &= -e_c^\mu b_\mu \delta_0^a - e_d^\nu b_\nu \delta_c^a \end{aligned} \right\}$$

...  $\square$

\* One can check that the conf. trans. are consistent with this expansion of the connection 2-form  $\omega_\mu^{ab}$ .

• We add a new constraint to fix the Gauge fields of SCT's. The following Gauge fixing allows us to solve for  $f_\mu^a$  algebraically.

$$e_b^\nu R_{\mu\nu}(M^{ab}) = 0, \quad R_{\mu\nu}(M^{ab}) = R_{\mu\nu}^{ab} + 8 f_\mu^{[a} e_{\nu]}^{b]}$$

Although,  $R_{\mu\nu}^{ab}$  contains  $b_\mu$  via  $\omega_\mu^{ab}$ , we will choose a gauge that removes this  $\Rightarrow (R_{\mu\nu})_{ab}$  is simply that of the Poincaré Curvature.

$$\begin{aligned} R_{\mu\nu ab} &= \partial_\mu \omega_{\nu ab} - \partial_\nu \omega_{\mu ab} + \omega_{\mu ac} \omega_\nu^{cb} \\ &= \omega_{\nu ac} \omega_\mu^{cb} \end{aligned}$$



$$2(D-2)f_m^a = -R_m^a + 2(D-2)e_m^a R$$

where

$$R_{mv} \equiv R_{\mu\nu} \quad e_a^\mu e_\nu^b, R \equiv R_\mu{}^\mu$$

Proof:  $e_b^\nu [R_{\mu\nu}^{ab} + 2(f_\mu^a e_\nu^b - f_\nu^a e_\mu^b)] = 0$   
 $\Rightarrow e_b^\nu [R_{\mu\nu}^{ab} + 2(f_\mu^a e_\nu^b - f_\nu^a e_\mu^b)]$   
 $R_\mu^a + 2(f_\mu^a D - f_\mu^a - f_\mu^a + e_b^\nu f_\nu^b e_\mu^a) = 0$   
 $R_\mu^a + 2(D-2)f_\mu^a + 2e_b^\nu f_\nu^b e_\mu^a = 0$  ★  
 $\Rightarrow e_a^\mu R_\mu^a + 2(D-2)e_a^\sigma f_\sigma^a + 2e_a^\sigma f_\sigma^a D = 0$   
 $R + (4D-4)e_a^\sigma f_\sigma^a = 0$

$$\Rightarrow e_a^\sigma f_\sigma^a = -\frac{1}{4(D-2)}R$$

Plugging this into ★  
 $R_\mu^a + 2(D-2)f_\mu^a + 2e_\mu^a \left(-\frac{1}{4(D-2)}R\right) = 0$

$$\therefore 2(D-2)f_\mu^a = -R_\mu^a + \frac{1}{2(D-2)}e_\mu^a R$$

\* With this Gauge choice, all the fields which are left are  $e_\mu^a$  vielbein & the gauge field dilatations  $b_\mu$ .

→ Impose conditions on  $b_\mu$  to fix the gauge of SCT's.

⑨

The trans.  $\boxed{\partial b_\mu = \partial_\mu \lambda_D + \epsilon \lambda_K - \eta_\mu}$

The  $\epsilon$  term can be written as

$$\delta_K(\lambda_K) b_\mu = 2(\lambda_K)_\mu$$

This implies that the values of this field can be changed at will by a SCT.

i.e.  $\exists (\lambda_K)_\mu$  s.t. we move  $b_\mu \rightarrow 0$ .

K-gauge:  $\boxed{b_\mu = 0}$

After this choice, only the vielbein remains & we have a purely gravitationally dynamic theory.

$$\boxed{(\lambda_K)_\mu = -\frac{1}{2} \partial_\mu \lambda_D}$$

This type of relation, imposed by preservation of a gauge condition is called a Decomposition law.

It expresses a 'Gauge-fixed' symmetry parameter in terms of Gauge Parameters of symmetries that remain in the theory.

The Action:

We will prove that the Conformally fixed



for a primary scalar field. of weight  $h$   
→ We couple the Conformal Gauge fields  
to a real scalar field.

Recall,

- (i) The Gauge trans. of a covariant quant.  
does not involve derivatives of the  
Symmetry parameters.
- (ii) If the alg. closes on the fields, then the  
trans. of a cov. quantity is a covariant quant.  
i.e. gauge fields only appear included either  
in cov. Derivatives  $D_\mu$  or in curvatures  
 $R_{ab}$ .

\* We can use these to calculate covariant  
quantities:

Scalar:  $\phi$ , Gauge trans. are (get)  $\phi \rightarrow \phi$   
Since  $\phi = h \ln \phi$ ,  $h$  to be determined.

Def: We define the Conformal d'Alembertian  
$$\square^c \phi := \eta^{ab} D_a D_b \phi$$

This is defined wrt.  $D_a$  instead of  $\partial_a$   
to have Gauge-invariance.

\* Recall, that

Which is a covariant quantity: its transformation does not involve derivatives of gauge parameters.

→ To calculate its transf. we can delete all terms that contain any gauge field other than  $e_\mu^a$ .

Relevant terms:

$$\left\{ \begin{array}{l} \delta e_\mu^a = -\lambda^{ab} e_{\mu b} - \lambda_D e_\mu^a \\ \delta \omega_\mu^{ab} = \dots - 4\lambda_\mu^{ca} e_\mu^b \\ \delta b_\mu = \dots + 2(\lambda_\mu)_c e_\mu^a \\ \delta f_\mu^a = \dots \end{array} \right\}$$

$$\begin{aligned} \delta(D_\mu \phi) &= \delta e_\mu^a (\partial_\mu - h b_\mu) \phi + e_\mu^a \delta(\partial_\mu - h b_\mu) \phi \\ &= f_\mu^a (\partial_\mu - h b_\mu) \phi \\ &\quad + e_\mu^a (h+1) \lambda_D \partial_\mu \phi - h(2\lambda_\mu)_c e_\mu^a e_\mu^c \phi \\ &= h b_\mu \phi \end{aligned}$$

\* But  $\delta(e_\mu^a e_\mu^a) = 0$

$$\delta e_\mu^a e_\mu^a + e_\mu^a (-\lambda^{ab} e_{\mu b} - \lambda_D e_\mu^a) = 0$$

$$\delta e_\mu^a e_\mu^a - \lambda_\mu^a \delta e_\mu^a - \lambda_D \delta e_\mu^a = 0$$

$$e_\mu^a \delta e_\mu^a = (\lambda_\mu^a + \lambda_D \delta_\mu^a) e_\mu^a$$

$$\delta e_\mu^a = (\lambda_\mu^a + \lambda_D \delta_\mu^a) e_\mu^a$$

$$\Rightarrow \delta e_\mu^a = (\lambda_\mu^a + \lambda_D \delta_\mu^a) e_\mu^a$$



$$\delta(D_a \phi) = (\lambda_a^b + \lambda_D \delta_a^b) D_b \phi + e_a^\mu(h) \lambda_D \partial_\mu \phi - h \lambda_a^b e_b^\mu e_a^\mu \phi - h b_\mu \lambda_D h \phi e_a^\mu$$

$$= -\lambda_a^b D_b \phi + \lambda_D D_a \phi - 2h(\lambda_a) a \phi + e_a^\mu h \lambda_D (\partial_\mu \phi - h b_\mu \phi)$$

$$\delta(D_a \phi) = -\lambda_a^b D_b \phi + (h+2)\lambda_D D_a \phi - 2h(\lambda_a) a \phi$$

With this, we can expand the cent.  $\square^c \phi$

How? we simply replace the coeff.

By their respective generators

$$\begin{aligned} \square^c \phi &= \eta^{ab} D_b D_a \phi \\ &= \eta^{ab} e_b^\mu (\partial_\mu D_a \phi - \delta(B_a^a D_a \phi)) \end{aligned}$$

Recall  $\lambda^{ab} \rightarrow \omega_{\mu}^{ab}$ ,  $\lambda_D \rightarrow b_\mu$ ,  $(\lambda_a)^a \rightarrow f_\mu^a$

$$\square^c \phi = \eta^{ab} e_b^\mu \left[ \partial_\mu D_a \phi + \omega_{\mu a}^b D_b \phi - (h+1)b_\mu D_a \phi + 2h f_{\mu a}^a \phi \right]$$

$$\square^c \phi = e^\mu_a (\partial_\mu D_a \phi - (h+1)b_\mu D_a \phi + \omega_{\mu a}^b D_b \phi + 2h f_{\mu a}^a \phi)$$

As above, we find

$D_a \phi$

$$\delta(\square^c \phi) = (h+2)\lambda_D \square^c \phi + (2D - 4h - 4)\lambda_a^a$$

$$\delta(\square^c \phi) = \text{cent.} + (h+2)\lambda_D \square^c \phi + (2D - 4h - 4)\lambda_a^a$$

Raises the weight by  $\frac{1}{2}$   
 we can make this a purely wgt  
 trans. By imposing  
 $2D - 4h - 4 = 0 \Leftrightarrow h = \left(\frac{1}{2}D\right) - 1$

\* For  $D=4 \Rightarrow h=1$ , which is  
 a (vector?). Thus, since  $\det(e^a_\mu) \in e$   
 has Kant. weight  $-D$ , since it is a  
 product of  $D$  factors of  $e^a_\mu$ .

$\therefore$

$$\delta_D(e\phi \square^c \phi) = -D \lambda_D(\cdot) + h \lambda_D(\cdot) + (h+2)(\cdot)$$

$$= \lambda_D(e\phi \square^c \phi) (-D + \frac{1}{2}D - 1 + \frac{1}{2}D + 1)$$

$$= 0$$

$\therefore$   $e\phi \square^c \phi$  

This is invariant under local dilations

$\therefore$   $\mathcal{I} = -\frac{1}{2} \int d^D x \ e\phi \square^c \phi$  (conf. inv. Action)

\* Now impose  $K$ -gauge Condition:  $\phi_\mu = 0$   
 and substitute  
 $e^a_\mu f^{\mu}_\nu = -\frac{1}{4(D-1)} R$

$$\Rightarrow \square^c \phi = e^a_\mu \left( \partial_\mu (e^{\nu}_a \partial^{\mu} \phi) + W_{\mu a b} e^b_\nu \partial^{\mu} \phi \right)$$

$$2) \left( \frac{2}{D-1} \right) R$$



$$\square^c \phi = e^{am} (\partial_m (e_a^v \partial_v \phi)) + e^{am} \omega_{mas} e_r^b \partial_v \phi$$

$$- 2 \left( \frac{1}{2} D - 1 \right) \frac{2}{4(D-1)} R$$

$$\therefore T = -\frac{1}{2} \int d^D x e \phi \left\{ e^{am} (\partial_m (e_a^v \partial_v \phi)) + \omega_{mas} e_r^b \partial_v \phi \right\}$$

$$- \frac{D-2}{4(D-1)} e \phi R$$

Must integrate by parts. Not sure this is worth it.....

$$\textcircled{a}: \int d^D x (\partial_m (e \phi e^{am})) e_a^v \partial_v \phi$$

$$= - \int d^D x e_b^v \partial_m e_b^v \phi e^{am} e_a^\sigma \partial_\sigma \phi e$$

$$+ \underline{e \partial_m \phi g^{uv} \partial_v \phi} + e \phi \partial_m (e^{am}) e_a^v \partial_v \phi$$

$$e_b^v \partial_m e_b^v \phi e^{am} e_a^\sigma \partial_\sigma \phi e + e \phi \partial_m (e^{am}) e_a^v \partial_v \phi$$

$$e_b^v \partial_m e_b^v \phi g^{uv} \partial_v \phi e + e \phi \omega^{as} (\partial_m e_b^u) e_a^v \partial_v \phi$$

$$e_b^\sigma \partial_m e_b^\sigma \phi g^{uv} \partial_v \phi e + e \phi \omega^{as} (\partial_m e_b^u) e_a^v \partial_v \phi$$

$$= (e \partial_\mu \phi) (e_b^\sigma \partial_m e_b^\sigma g^{uv} + \omega^{as} (\partial_m e_b^u) e_a^v)$$

$$\left\{ \begin{aligned} e_b^c e_a^v \omega^{as} e_b^\sigma &= g^{\sigma v} e_a^c \\ \omega_{ce} \omega^{cb} e_b^\sigma &= g^{\sigma v} e_b^c \omega_{ce} \\ e_a^\sigma &= g^{\sigma v} e_b^c \omega_{ce} \\ e_a^v &= g^{\sigma v} e_b^c \omega_{ca} \end{aligned} \right.$$

$$(e \partial_\mu \phi) (e_b^\sigma \partial_m e_b^\sigma g^{uv} + \omega^{as} \partial_m (e_b^u) g^{v\sigma} e_a^c \omega_{ce})$$

$$\partial_m (e_b^u) g^{v\sigma} e_b^\sigma$$

$$g^{\sigma\lambda} e_b^c \omega_{cb} \partial_m e_b^\sigma g^{uv}$$



\* These terms will cancel w/ the expansion of  $\square$  & yield  $R\phi^2$

$$\square : \frac{1}{2} \int d^D x e \partial_\mu \phi g^{\mu\nu} \partial_\nu \phi$$

yields the scalar part. of the action

Do this tomorrow

$$I = \int d^D x e \left( \frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) + \frac{D-2}{2(D-1)k} R \phi^2 \right)$$

The final gauge fixing  $\phi = \sqrt{\frac{4(D-1)}{(D-2)k}}$  yields the Einstein-Hilbert Action

→ The scalar field  $\phi$  is not physical and is not a physical field of the theory.

Homothetic Killing Vectors:

→ Look at a system of scalars  $\phi^i(x)$  whose dynamics are described by a nonlinear  $\sigma$ -model w/ target space Mink metric  $g_{ij}(\phi)$ .

$$\phi : M^D \rightarrow M^D, x^\mu \mapsto \phi^i(x^\mu)$$

$\Rightarrow \int d^D x e \frac{1}{2} g^{ij}(\phi) \partial_\mu \phi_i \partial^\mu \phi_j$  w/ the new added Conf. term.  $\sim R\phi^2$ .



\* Vector fields that generate the symms

$\xi^\mu(x)$ : holomorphic Killing vectors

$\hookrightarrow$  AKA. Closed & exact homothetic Killing vect.

\* When the target space has non-trivial topology, we cannot have (in general)

$$\delta_D \phi = \lambda \phi^i \lambda_i$$

Instead, we have (Dilations)

$$\delta_D \phi^i = \lambda_D K_D^i(\phi)$$

$K_D^i(\phi) \equiv$  (homothetic Killing vector)

which also plays a role in SCTs.

Globally: We find  $\lambda_D(x) = \lambda_D = 2x \cdot \lambda_k$

Locally: we get a mixing from the gen.  $b_\mu$ , which transforms under SCT's.

Global SCT:

$$\delta_{\text{SCT}} \phi = (x^2 \lambda_k^m - 2x^m (x \cdot \lambda_k)) \partial_\mu \phi^i - 2x^\mu \lambda_k K_\mu^i(\phi)$$

→ why?: Recall

$$\delta \phi = \xi^\mu \partial_\mu \phi + \dots + \lambda_k^m K_\mu^i(\phi) x^\mu$$

$\xi^\mu$  contains the coordinate SCT term

$$(x^2 \lambda_k^m - 2x^m (x \cdot \lambda_k))$$

Basically looking at  $\delta \phi$  and dropping all terms which do not contain  $\lambda_k$ .

• We have also assumed  $K_\mu^i(\phi) = 0$

→ why???????

↳ This (must)? mean that we are assuming  
SCT's do not act or alter the  
rep that  $q$  is in  $Q$ : Would this  
not imply that the term coming  
from  $\Delta_0(x)$  is purely coordinate:  
Yes,  $\Delta_0(x)$  is the rescaling conf.  
factor of  $ds^2 = g_{ab} dx^a dx^b$ .

\* Assumption valid if there are no lower  
conf. weight scalars in the theory  
→ ahhh: Recall that  $K_\mu \phi^i \propto (h-1)\chi^i$   
lowers the conf. weight by 1.

\* Non linear  $\sigma$ -model is invariant if  
 $K_\mu^i$  is closed homothetic Killing vector  
(proof below)

The (extended) non linear  $\sigma$ -model is also  
invariant under SCT's iff  $K_\mu^i$  is a closed  
homothetic.

$$K_\mu^i \in \Gamma(TM^D)$$

→ This vector is needed to  
Rescale conf. Gramm. of the



# A nonlinear $\sigma$ -model field theory

\* Let  $M^D$  be the target mfd w/ real coord's  $\phi^i$  & metric  $g_{ij}(\phi)$

Killing vector:  $\phi^i \mapsto \phi^i + \epsilon \tilde{K}^i(\phi)$   
 $\Leftrightarrow \mathcal{L}_{\tilde{K}} g_{ij} = 0$

Isometries:  $f: M \rightarrow M$

$$f^* g_{f(P)} = g_P$$

$$\left\{ \begin{aligned} f^* g_{f(P)}(X, Y) &= g_P(X, Y) = g_{f(P)}(f_* X, f_* Y) \\ \text{If } X &= X^\mu \frac{\partial}{\partial x^\mu} \Rightarrow f_* X = X^\mu \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial}{\partial y^\alpha} \\ \text{where } y^\alpha &= \phi^\alpha(f(P)) \text{ on } (U, \phi) \end{aligned} \right\}$$

$$\Rightarrow g_{\alpha\beta}(f(P)) \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial y^\beta}{\partial x^\nu} = g_{\mu\nu}(P)$$

Conformal Killing Vectors:  $f: M^D \rightarrow M^D$

$$\text{under } (f^* g)_{ij} \equiv e^{2\sigma(\phi)} g_{ij}$$

$$\text{with } (f(\phi))^i := \phi^i + \epsilon \tilde{K}^i$$

Thus,

$$\begin{aligned} & g_{kl}(\phi + \epsilon \tilde{K}) \left( \frac{\partial(\phi^k + \epsilon \tilde{K}^k)}{\partial \phi^i} \right) \left( \frac{\partial(\phi^l + \epsilon \tilde{K}^l)}{\partial \phi^j} \right) = e^{2\sigma(\phi)} g_{ij} \\ & (g_{kl}(\phi) + \epsilon \tilde{K}^i \partial_i g_{kl}(\phi)) (\delta_i^k + \epsilon \partial_i \tilde{K}^k) (\delta_j^l + \epsilon \partial_j \tilde{K}^l) \\ & = e^{-2\sigma(\phi)} g_{ij} \\ \text{LHS} &= (g_{ik} + \epsilon \partial_i \tilde{K}^k g_{kl} + \epsilon \tilde{K}^i \partial_i g_{kl}) (\delta_j^k + \epsilon \partial_j \tilde{K}^k) \end{aligned}$$

$$\begin{aligned}
 &= g_{ij} + \epsilon (\partial_i k^k g_{kj} + \partial_k k^i g_{ij} + g_{ik} \partial_j k^k) \\
 &= g_{ij} + \epsilon \mathcal{L}_k g_{ij}
 \end{aligned}$$

$$RHS = g_{ij} + \sigma(\phi) g_{ij}$$

Since  $\sigma(\phi)$  is an infinitesimal in  $\epsilon$   
 $\sigma(\phi) := \epsilon h(\phi)$

$$(\mathcal{L}_k g)_{ij} = 2\nabla_{[i} k_{j]} = h(\phi) g_{ij}$$

(conf. Killing Vector)

Note, we can solve for  $h(\phi)$ :

$$k^m \partial_m g_{ij} + \partial_i k^m g_{mj} + \partial_j k^m g_{im} = h(\phi) g_{ij}$$

$$g^{ij} k^m \partial_m g_{ij} + 2 \partial_m k^m = h(\phi) D$$

$$h(\phi) = \left(\frac{1}{D}\right) (g^{ij} k^m \partial_m g_{ij} + 2 \partial_m k^m)$$

Homothetic Killing vector:  $h(\phi) := 2h$

$$(\mathcal{L}_k g)_{ij} = \nabla_i k_j + \nabla_j k_i = 2h g_{ij}, \quad h \in \mathbb{R}$$

Let  $K_i \equiv g_{ij} K^j$ , then the one form  
 $K \in \Omega^1(M^D)$ ,  $K = K^i g_{ij} d\phi^j$

Closed Homothetic Killing Vector:

The Killing form  $K$  is Closed i.e.

$$dK = 0 \Leftrightarrow \partial_i K_j - \partial_j K_i = 0$$

$$\nabla_i K_j = h g_{ij}$$



Conditions for closed Kill. vec.:

locally  $\exists$  a potential  $K(\phi)$  on  $M^D$  s.t.

$$K = dK = \frac{\partial K}{\partial \phi^i} d\phi^i$$

$$\Rightarrow [K_i = K^j g_{ij} = \partial_i K]$$

$$\therefore \nabla_i (\partial_j K) = g_{ij} \quad (\text{Closed homothetic Killing vect.})$$

Integrability conditions:

$$[\nabla_i, \nabla_j] K_\alpha = R_{ijkl} K^l = 0$$

\* This strongly restricts the target space which support SCT's.

Proof:

$$[K_i, K_j] K_\alpha = (\nabla_i \nabla_j - \nabla_j \nabla_i) K_\alpha = \nabla_i (h g_{j\alpha}) - \nabla_j (h g_{i\alpha})$$

$$\text{but } \nabla_a g_{bc} = 0 \quad \text{so}$$

$$[\nabla_i, \nabla_j] K_\alpha = 0 \Leftrightarrow R_{ijkl} K^l = 0 \quad \square$$

\* Define  $K_j := K_{Dj}$ . Then the (CHKV) eqn becomes (expanding  $\nabla_i$ )

$$g^{ik} \nabla_i K_{Dj} = h \delta_i^j = \nabla_i K_D^j$$

$$\Leftrightarrow \nabla_i h^j = h \delta_i^j$$

$$(\partial_i K_D^j + \Gamma_{ik}^j K_D^k) = h \delta_i^j$$

when the geometry is trivial &  $\Gamma_{ik}^j = 0$

we find  $\partial_i K_D^j = h \delta_i^j \Leftrightarrow K_D^i = h \phi^i$ ,

which was our initial assumption.

## The nonlinear $\sigma$ -model in $1M^D$ :

• Global Dilations & sets of scalar fields

$$\delta \phi^i = \xi^\mu \partial_\mu \phi^i + \Delta_D(x) \kappa_D^i(\phi)$$

Since  $\kappa_D^i(\phi) = 0$

• Note since  $\Delta_D(x) = \frac{1}{D} \partial_\rho x^\rho$

$$\Leftrightarrow [K_{\mu\nu} \Delta_D(x) = \partial_\mu \xi_\nu]$$

Then,  $\mathcal{L} = -\frac{1}{2} g_{ij}(\phi) \partial_\mu \phi^i \partial^\mu \phi^j$

transforms as

$$\delta \mathcal{L} = \partial_\mu (\xi^\mu \mathcal{L}) + (-D + 2 + 2h) \mathcal{L} \Delta_D(x) - \kappa_{Di} \partial_\mu \phi^i \partial^\mu \phi^j$$

iff  $\kappa_D^i$  are  $(h, 0)$  of weight  $h$ .

(i) Then with choice of weight  $h = \frac{D}{2} - 1$

there is dilation invariance

$\rightarrow$  i.e. invariance of the action under

$$\Delta_D = \Delta_D.$$

iii) If  $\kappa_{Di} = \partial_i K(\phi)$ , i.e.  $(0, h)$  the action is also invariant for  $\Delta_D = -2(x \cdot \partial_x)$

proof:

$$\begin{aligned} \delta \mathcal{L} &= -\frac{1}{2} \delta g_{ij} \partial_\mu \phi^i \partial^\mu \phi^j - \frac{1}{2} g_{ij} \delta (\partial_\mu \phi^i) \partial^\mu \phi^j \\ &\quad - \frac{1}{2} g_{ij} \partial_\mu \phi^i \delta (\partial^\mu \phi^j) \\ &= \Delta_D g_{ij} \partial_\mu \phi^i \partial^\mu \phi^j \end{aligned}$$

$$\begin{aligned} * \partial_\mu (\xi^\nu \partial_\nu \phi^i + \Delta_D(x) \kappa_D^i) \\ = \partial_\mu \xi^\nu \partial_\nu \phi^i + (\partial_\mu \partial_\nu \phi^i) \xi^\nu + \partial_\mu \Delta_D \kappa_D^i + \Delta_D \partial_\mu \kappa_D^i \end{aligned}$$

$\Rightarrow$



$$\begin{aligned}
 & \times \partial^\mu \phi^j \\
 & - g_{ij} \partial_\mu \xi^\nu \partial_\nu \phi^i \partial^\mu \phi^j - g_{ij} \partial_\mu \partial_\nu \phi^i \partial^\mu \phi^j \xi^\nu \\
 & - g_{ij} (\partial_\mu \Lambda_0) \kappa^i_j \partial^\mu \phi^j - g_{ij} \Lambda_0 \partial_\mu \kappa^i_j \partial^\mu \phi^j
 \end{aligned}$$

→ last term ✓

$$\begin{aligned}
 (\partial_\mu \xi_\nu) \partial^\nu \phi^i \partial^\mu \phi^i &= \left( \frac{2}{D} \eta_{\mu\nu} (\partial_\rho \xi^\rho) - \partial_\nu \xi_\mu \right) \partial^\nu \phi^i \partial^\mu \phi^i \\
 &= \frac{2}{D} (\partial_\rho \xi^\rho) \partial_\mu \phi^i \partial^\mu \phi^i g_{ij} - \partial_\nu \xi_\mu \partial^\nu \phi^i \partial^\mu \phi^i g_{ij} \\
 \therefore \partial_\mu \xi_\nu \partial^\nu \phi^i \partial^\mu \phi^i &= \frac{2}{D} (\partial_\rho \xi^\rho) \partial_\mu \phi^i \partial^\mu \phi^i g_{ij}
 \end{aligned}$$

\* Better. Suppose  $\delta \mathcal{L} = \partial_\mu (\xi^\mu \mathcal{L}) + \dots$   
 $\partial_\mu \xi^\mu \mathcal{L} + \xi^\mu \partial_\mu \mathcal{L} + \dots$

$$\text{But } \partial_\mu (\xi^\mu) = \frac{1}{D} \eta_{\mu\nu} (\partial_\mu \xi^\mu) = \eta_{\mu\nu} \Lambda_0$$

$$\partial_\mu \xi^\mu = D \Lambda_0$$

$$\begin{aligned}
 \Rightarrow D \Lambda_0 \mathcal{L} + \xi^\mu \partial_\mu \mathcal{L} &= D \mathcal{L} \Lambda_0 + 2(1+h) \mathcal{L} \Lambda_0 \\
 &\quad - \kappa_{0i} \partial_\mu \phi^i \partial^\mu \Lambda_0 \\
 &= \xi^\mu \partial_\mu \mathcal{L} + 2(1+h) \mathcal{L} \Lambda_0 - \kappa_{0i} \partial_\mu \phi^i \partial^\mu \Lambda_0
 \end{aligned}$$

$$?: \text{ Should } \delta g_{ij} = \frac{\partial g_{ij}}{\partial \phi^a} \delta \phi^a ?$$

± If so, then we obtain:

$$\begin{aligned}
 \delta \mathcal{L} &= -\frac{1}{2} (\partial_\mu g_{ij} (\xi^\sigma \partial_\sigma \phi^a + \Lambda_0 \kappa^a_{0i}(\phi))) \partial_\mu \phi^i \partial^\mu \phi^j \\
 &= -\frac{1}{2} (\partial_\mu g_{ij} \xi^\sigma \partial_\sigma \phi^a + \Lambda_0 \kappa^a_{0i}(\phi)) \partial_\mu \phi^i \partial^\mu \phi^j
 \end{aligned}$$

$$\rightarrow g_{ij} \xi^\sigma \partial_\sigma \partial_\mu \phi^i \partial^\mu \phi^j$$

$$= -\frac{1}{2} \partial_\mu g_{ij} \xi^\sigma \partial_\sigma \phi^a \partial_\mu \phi^i \partial^\mu \phi^j + \text{irr.}$$

$$- g_{ij} \xi^\sigma \partial_\sigma \partial_\mu \phi^i \partial^\mu \phi^j$$



$$\begin{aligned}
 &= \xi^\sigma \partial_\sigma (\mathcal{L}) + \dots \\
 &= \partial_\mu (\xi^\mu \mathcal{L}) - \partial_\mu \xi^\mu \mathcal{L} + \text{irr.} \\
 &= \partial_\mu (\xi^\mu \mathcal{L}) - D \Lambda_0 \mathcal{L} - \frac{1}{2} \partial_\alpha g_{ij} \Lambda_0 k_\alpha^i (\phi) \partial_\mu \phi^j \partial^\mu \phi^k
 \end{aligned}$$

$$\partial_\mu (\xi^\sigma \partial_\sigma \phi^i + \Lambda_0 \alpha) k_\sigma^i (\phi)$$

$$\begin{aligned}
 &= (\partial_\mu \xi^\sigma \partial^\sigma \phi + \xi^\sigma \partial_\mu \partial_\sigma \phi + \partial_\mu \Lambda_0 \alpha) k_\sigma^i \\
 &\quad + \Lambda_0 \partial_\mu \phi^k \partial_\alpha k_\sigma^i (\phi)
 \end{aligned}$$

$$\left[ -g_{ij} (\partial_\mu \xi^\sigma \partial^\sigma \phi + \partial_\mu \Lambda_0 \alpha) k_\sigma^i + \Lambda_0 \partial_\mu \phi^k \partial_\alpha k_\sigma^i \right] \times \partial^\mu \phi^j$$

$$\begin{aligned}
 &\rightarrow -g_{ij} \partial_\mu \xi^\sigma \partial^\sigma \phi^i \partial^\mu \phi^j - g_{ij} (\partial_\mu \Lambda_0) k_\sigma^i \partial^\mu \phi^j \\
 &\quad - g_{ij} \Lambda_0 \partial_\mu \phi^k \partial_\alpha k_\sigma^i \partial^\mu \phi^j
 \end{aligned}$$

\* Ignore — as it is the final term:

$$\begin{aligned}
 \therefore &-g_{ij} \partial_\mu \xi^\sigma \partial^\sigma \phi^i \partial^\mu \phi^j - g_{ij} \Lambda_0 \partial_\mu \phi^k \partial_\alpha k_\sigma^i \\
 &= -g_{ij} \partial_\mu \xi^\sigma \Lambda_0 \partial^\sigma \phi^i \partial^\mu \phi^j - g_{ij} \Lambda_0 \partial_\mu \phi^k \partial_\alpha k_\sigma^i \partial^\mu \phi^j \\
 &= -g_{ij} \Lambda_0 \partial_\mu \phi^i \partial^\mu \phi^j - g_{ij} \Lambda_0 \partial_\mu \phi^k \partial^\mu \phi^j (\partial_\alpha k_\sigma^i)
 \end{aligned}$$

Assumption:  $k_\sigma^i$  is homogeneous w/ weight

$$h \Rightarrow \partial_i k_\sigma^i + \partial_\sigma k_\sigma^i = 2h g_{ij}$$

$$\begin{aligned}
 &\rightarrow -g_{ij} \Lambda_0 \partial_\mu \phi^i \partial^\mu \phi^j - g_{ij} \Lambda_0 \partial_\mu \phi^k \partial^\mu \phi^j h g^{ki} \\
 &= \Lambda_0 [(-g_{ij} \partial_\mu \phi^i \partial^\mu \phi^j) (1 + h)]
 \end{aligned}$$

helllll yeah...

$$= \Lambda (2h + 2) \mathcal{L}$$



(ii) For weights  $h = \frac{D}{2} - 1$

$(0 + D - 2 + 2) = 0$  middle term,

- This is, as we found above, dilation invariance:  $\Delta_D = \lambda_D$  so last term vanishes as well

$$\delta \mathcal{L} = \partial_\mu (\xi^\mu \mathcal{L}) \quad \checkmark$$

→ If  $K_0(\phi) = \partial_i K(\phi)$ , then the action is invariant under  $\Delta_D = -2x \cdot \partial_a$   
 $\square$

### Coupling the $\sigma$ -model to Gravity:

We now couple  $\sigma$  to gravity & consider its local spacetime-symmetries: Dilations & SCT's each w/ local parameters  $\lambda(x)$  &  $\lambda_\mu^a(x)$ .

$$\delta \phi^i = \lambda_D \phi^i(\phi), \quad \delta e_\mu^a(x) = -\lambda_D e_\mu^a$$

$$\delta b_\mu = \partial_\mu \lambda_D + 2 \lambda_\mu^a e_{\mu a}$$

$$\delta f_a^i = 2 \lambda_D f_a^i + e_\mu^i (\partial_\mu - b_\mu) \lambda_\mu^a + e_\mu^i \omega_\mu^{ab} (e, b) \lambda_{\mu b}$$

Here we are assuming  $\delta_\mu \phi = \partial_\mu \phi = 0$

since the theory does not have any fields w/ zero conformal weight:  $h=0$ .

Same for  $e_\mu^a$ . The theory then has

H...

$$L := -\frac{1}{2} e g_{ij}(\phi) D_a \phi^i D^a \phi^j + \frac{e}{h} f_a^g K_D^i g_{ij} K_D^j$$

where  $D_a \phi := e_a^\mu (\partial_\mu \phi^i - b_{\mu}^i)$

$\sigma$ -model-gravity:

$$S = -\frac{1}{2} \int d^D x \left( e g_{ij}(\phi) D_a \phi^i D^a \phi^j + \frac{e}{h} f_a^g K_D^i g_{ij} K_D^j \right)$$

We then calculate the variation of this Action to better understand when this theory is invariant under  $\text{Conf}(\mathcal{M}^D)$ .

Derivation of ~~~~~: We had  $R\phi^2$  and

we have  $f_a^g = -\frac{1}{4(D-1)} R \Rightarrow R = -4 f_a^g (D-1)$

$$\Rightarrow \frac{(D-2)}{4(D-1)} (4 f_a^g) (D-1) \phi^2 = -(D-1) f_a^g \phi^2 = -h f_a^g \phi^2$$

Recall  $\phi^2 \rightarrow \phi^i g_{ij} \phi^j$

$$\therefore -h f_a^g \phi^i g_{ij} \phi^j \xrightarrow{\text{re}} \text{}$$

Q: where did  $R\phi^2$  come from?

In our uncoupled theory we had

$$D_a \phi^i = e_a^\mu (\partial_\mu \phi^i - b_{\mu}^i)$$

inside of  $D^c \phi^i := \eta^{ab} D_a D_b \phi^i$

We NOW have  $D_a \phi^i := e_a^\mu (\partial_\mu \phi^i - b_{\mu}^i K_D^i)$

So, we can make the appropriate

substitution  $(h \phi^i) g_{ij} (h \phi^j) \approx K_D^i g_{ij} K_D^j$

or  $\phi^i g_{ij} \phi^j \approx \frac{1}{h^2} K_D^i g_{ij} K_D^j$

Plus we have  $\text{~~~~~}$  and



$$\text{wavy line} = \frac{e}{\hbar} f_a^a \kappa_D^i g_{ij} \kappa_D^j$$

\* Showing invariance of this Action:

$$① = -\frac{1}{2} e g_{ij}(\phi) (e_a^\mu (\partial_\mu \phi^i - b_\mu \kappa_D^i)) (e^{\nu a} (\partial_\nu \phi^j - b_\nu \kappa_D^j))$$

Ignore the  $(-1/2)$  factor for now:

$$\begin{aligned} \delta(\mathcal{L}) &= \delta e g_{ij} D_\mu \phi^i D^\mu \phi^j + e (\delta g_{ij}) D_\mu \phi^i D^\mu \phi^j \\ &\quad + 2 e g_{ij} \delta(D_\mu \phi^i) D^\mu \phi^j \\ &= -\lambda_D e e_a^\mu e_\mu^a g_{ij} D_\mu \phi^i D^\mu \phi^j - 2 e \lambda_D D_\mu \phi^i D^\mu \phi^j g_{ij} \\ &\quad + \dots \end{aligned}$$

$$\begin{aligned} \bullet \delta(e_a^\mu (\partial_\mu \phi^i - b_\mu \kappa_D^i)) &= -\lambda_D D_a \phi^i + e_a^\mu (\lambda_D \partial_\mu \kappa_D^i - 2 \lambda_\kappa e_{\mu b} \kappa_D^i) \\ &\quad - e_a^\mu \partial_\mu \kappa_D^i \lambda_D \kappa_D^a(\phi) b_\mu \\ &= -\lambda_D D_a \phi^i + e_a^\mu \lambda_D \partial_\mu \phi^k \partial_\mu \kappa_D^i - 2 \lambda_\kappa e_a^\mu \kappa_D^i \\ &\quad - \lambda_D e_a^\mu \kappa_D^k \partial_\mu \kappa_D^i b_\mu \\ &= -\lambda_D D_a \phi^i + \lambda_D (e_a^\mu) (\partial_\mu \kappa_D^i) (\partial_\mu \phi^k - b_\mu \kappa_D^i) \\ &\quad - 2 \lambda_\kappa D \kappa_D^i \\ &= -\lambda_D D_a \phi^i + \lambda_D (\partial_\mu \kappa_D^i) D_a \phi^k - 2 \lambda_\kappa \kappa_D^i \\ \therefore \delta(D_a \phi^i) &= \lambda_D (-D_a \phi^i + \partial_\mu \kappa_D^i D_a \phi^\mu) - 2 \lambda_\kappa \kappa_D^i \end{aligned}$$

$$\begin{aligned} \rightarrow \delta(\mathcal{L}) &= -\lambda_D \textcircled{1} D - 2 \lambda_D \textcircled{2} \\ &\quad + 2 e g_{ij} [(\lambda_D) (-D_a \phi^i + \partial_\mu \kappa_D^i D_a \phi^\mu) - 2 \lambda_\kappa \kappa_D^i] D^\mu \phi^j \\ &= (-\lambda_D) (1+2) \textcircled{1} D - 2 \lambda_D \textcircled{2} + 2 \lambda_D \partial_\mu \kappa_D^i \dots D_a \phi^\mu D^\mu \phi^j \end{aligned}$$

$$\begin{aligned}
 &= 4(e g_{ij} (\lambda \kappa)_a k_b^i D^a \phi^b) \\
 &= (-\lambda_0) \left[ (D + 4)(e g_{ij} D^a \phi^i D^a \phi^j) - 2 \partial_a k_b^i g_{ij} D^a \phi^b D^a \phi^j \right] \\
 &= 4 e g_{ij} (\lambda \kappa)_a k_b^i D^a \phi^j \quad \checkmark
 \end{aligned}$$

We now take the variation of the second term

$$\textcircled{b} = \frac{e}{h} f_a^i k_b^j g_{ij} k_b^j$$

$$\begin{aligned}
 \delta(b) &= (-\lambda_0 D) \textcircled{b} + \frac{e}{h} \left( 2 \lambda_0 f_a^i + e_a^\mu \omega_\mu^{ab}(e) (\lambda \kappa)_b \right) k_b^i \\
 &\quad + \frac{e}{h} f_a^i \delta(k_b^j g_{ij} k_b^j)
 \end{aligned}$$

$$\begin{aligned}
 &2 \delta k_b^i g_{ij} k_b^j + k_b^i \delta g_{ij} k_b^j = 2 \partial_a k_b^i k_b^a g_{ij} k_b^j \lambda_0 \\
 &\quad + -2 k_b^i g_{ij} k_b^j \lambda_0 \\
 &= 2 g_{ij} k_b^j \lambda_0 (\partial_a k_b^i k_b^a - k_b^i)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{b} &= (-\lambda_0 D) \textcircled{b} + \textcircled{b} \overset{4?}{2 \lambda_0} + \frac{e}{h} e_a^\mu \omega_\mu^{ab}(e) (\lambda \kappa)_b k_b^i g_{ij} k_b^j \\
 &\quad + \frac{e}{h} f_a^i 2 g_{ij} k_b^j \lambda_0 \partial_a k_b^i k_b^a - 2 \lambda_0 \textcircled{b}
 \end{aligned}$$

$$\begin{aligned}
 &= (-\lambda_0) \left( \frac{e}{h} f_a^i k_b^j g_{ij} k_b^j \right) (D) \\
 &\quad + \frac{e}{h} k_b^i g_{ij} k_b^j (e_a^\mu \omega_\mu^{ab}(e) (\lambda \kappa)_b) + \frac{2e}{h} f_a^i g_{ij} k_b^j \lambda_0 \partial_a k_b^i k_b^a
 \end{aligned}$$

If we now assume that  $k_b^i$  is homothetic, then and  $h = \frac{p}{2} - 1$ , then the Dilation terms  $\propto \lambda_0(\cdot)$  vanish separately for both  $\textcircled{a}$  &  $\textcircled{b}$ .

(ii) For SCT's  $\sim (\lambda \kappa)_b$  we add both terms and rewrite the term in  $\textcircled{a}$  as



$$\partial_a (K^j_i g_{ij} K^i_j) = h_{ab} \partial_a \phi^b$$

to see that  $L$  is conformally inv!

Def: We define a closed homothetic killing vec.:

Let  $K \in \Gamma(TM^D)$  and  $Y \in \Gamma(TM^D)$

then  $K$  is (h.k.v.) if

$$\boxed{\nabla_Y K = Y} \quad \forall Y \in \Gamma(TM^D)$$

→ Proof of eq.:

$$\nabla_Y K = Y^i (\nabla_i K) = Y^i \partial_i$$

$$\Rightarrow \nabla_i K^j (\partial_j) = (\nabla_i K^j) \partial_j + K^j \nabla_i \partial_j = \nabla_i K^j \partial_j + K^j \Gamma_{ij}^k \partial_k$$

$$= \nabla_i K = (\partial_i K^k + \Gamma_{ij}^k K^j) \partial_k$$

$$\therefore \partial_i \stackrel{\wedge}{=} (\partial_i K^k + \Gamma_{ij}^k K^j) \partial_k$$

$$\Leftrightarrow \partial_i K^k + \Gamma_{ij}^k K^j = \delta_i^k$$





